Tali Gankin

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EDUCATION

University of Southern California, Viterbi School of Engineering - Los Angeles, CA **Expected Graduation: May 2028**
Bachelor of Science in Computer Engineering and Computer Science
Honors: Presidential Scholar Recipient, Viterbi Dean's List, Viterbi CURVE Research Fellowship
Relevant Coursework: Principles of Software Development, Algorithms and Theory of Computing, Distributed Systems and Internet of Things, Data Structures, OOP and Design, Discrete Methods in CS, Digital System Design with FPGAs

SKILLS

Languages: Python, C/C++, Java, JavaScript, Verilog, HTML/CSS
Tools & Technologies: Linux/Unix, Git, Docker, TCP/IP Networking, Raspberry Pi, SQL (familiarity), REST APIs, GitHub
Technical Concepts: Prompt Engineering, AI Workflow Optimization, API Integration, Agile Methodologies, Design Thinking

EXPERIENCE

Consider it Done Technologies (CIDT) **Chicago, IL**
Software Engineering Intern *May 2026 - August 2026*
- Tested API-driven search across 10K+ materials in Python, identifying issues and evaluating accuracy pre-deployment
- Prototyped a Python image-indexing workflow to map features to searchable keywords and reduce manual material lookup
- Collaborated with engineers, UI/UX designers, and Spec-ID stakeholders; used Figma/feedback to refine testing & usability

CURVE Fellow (Research) **Los Angeles, CA**
Research Assistant *January 2025 - May 2026*
- Developed Python/MATLAB scripts to analyze 10K+ environmental data points, improving processing efficiency by 20%
- Applied graph-based interpolation techniques to reconstruct missing values across large-scale environmental datasets
- Evaluated model performance across varying graph parameters using quantitative error metrics to assess overall robustness

PROJECTS

Gesture-Controlled IoT System **Spring 2026**
Embedded Systems / IoT Project
- Developed a Raspberry Pi system using computer vision to detect hand gestures and control hardware responses in real time
- Integrated Python image processing with GPIO-controlled hardware, connecting software classification to physical outputs
- Applied IoT and embedded concepts including Linux, networking, sensor interfacing, and hardware/software integration

Attendance Dashboard **Fall 2025**
Software Engineering Project
- Built a full-stack dashboard to ingest, process, and visualize CSV-based data for 60+ users using JavaScript and Python
- Designed a data pipeline integrating Google Sheets APIs, enabling real-time tracking and analysis of participation metrics
- Improved operational efficiency by automating data processing workflows, reducing manual tracking by ~2 hrs/week

BinBot **Fall 2024**
Technical Project Manager
- Developed a Python-based classifier to identify recyclable materials from sensor input data (~90% accuracy / 50+ trials)
- Analyzed multi-sensor input data to improve classification, system reliability, and overall decision-making accuracy
- Presented data visualizations built in Python and Excel to technical and non-technical audiences at a competition (1st place)

LEADERSHIP AND INVOLVEMENT

USC Theta Tau Engineering Society **Los Angeles, CA**
Scribe *May 2025 - August 2026*
- Managed sensitive organizational records for 40+ members and 100+ applicants, ensuring accuracy and confidentiality
- Automated attendance and member-tracking workflows, reducing work by ~2 hours/week & improving attendance tracking

Professional Chair and Social Chair *September 2024 - May 2025*
- Fostered a collaborative community of 70+ members through weekly professional workshops and networking opportunities
- Optimized allocation of $6,000+ in organizational funds to strategically boost events and strengthen member engagement

Joint Education Program - Peace Project **Los Angeles, CA**
Program Assistant *September 2025 - Present*
- Coordinate school partnerships, outreach, and program logistics connecting USC students with local K–12 classrooms